

CAAM/STAT31470: Homework #6

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: One of the most important tools in complex analysis is the Cauchy Residue Theorem, which allows you to compute many integrals by instead finding residues. Here you will practice finding Laurent series, residues and integrals. Note that it is important to always think about what contour you are integrating over when looking to compute a Laurent series or residue.

Homework is due: **Wednesday May 14, 11:59pm.**

Problem 6.1 (6 points)

We have seen that the function

$$f(z) = \frac{1}{(z-i)^2}$$

has a simple Laurent expansion centered around $z = i$. However, we can find many different expansions for f relevant on different subsets of $\mathbb{C}$. Find a convergent series representation (either Taylor or Laurent) centered around:

- (a) $z = 0$ which converges for z close to 0,
- (b) $z = 0$ for z far from zero;

and in each case state the radius of convergence.

Problem 6.2 (6 points)

Write the two Laurent series for the function

$$f(z) = \frac{1}{z(1+z^2)}$$

in powers of z and specify their domains of convergence. Using the residues of these series write down the values of the integral

$$\int_C f(z) \, dz = 0$$

and the simple contours C for which you get those values.

Problem 6.3 (6 points)

Compute the residue at $z = 0$ of the following functions.

(1) $(a) \quad f(z) = z \cos\left(\frac{1}{z}\right), \quad (b) \quad f(z) = \frac{\sinh z}{z^4(1-z^2)}$

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Problem 6.4 (6 points)

- (a) Let z be any complex number and C denote the unit circle $w = e^{i\phi}$ for $\phi \in [-\pi, \pi]$. Then using the formula for the coefficients of the Laurent series to show that

$$\exp\left[\frac{z}{2}\left(w - \frac{1}{w}\right)\right] = \sum_{n=-\infty}^{\infty} J_n(z)w^n$$

for all $0 < |w| < \infty$ where

$$J_n(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp(-i(n\phi - z \sin \phi)) d\phi$$

- (b) By using properties of even and odd functions show that the coefficients can be rewritten as

$$J_n(z) = \frac{1}{\pi} \int_0^{\pi} \cos(n\phi - z \sin \phi) d\phi$$

These coefficient functions are called Bessel functions of the first kind which appear all over applied mathematics. This calculation shows that we can use complex analysis to write these functions as a discrete Fourier transform.

Problem 6.5 (6 points)

Let C denote the circle $|z| = 1$, taken counterclockwise. Show that

$$\int_C \exp\left(z + \frac{1}{z}\right) dz = 2\pi i \sum_{n=0}^{\infty} \frac{1}{n!(n+1)!}$$

Hint: Find the Taylor series for e^z at $z = 0$ and integrate term-by-term to show

$$\int_C \exp\left(z + \frac{1}{z}\right) dz = \sum_{n=0}^{\infty} \frac{1}{n!} \int_C z^n \exp\left(\frac{1}{z}\right) dz.$$

End of homework